

C- program:

```
int a;  
int b;  
int result;  
a = 3;  
b = a + 2;  
result = a + b + 5;
```

Flattened IR

```
t1 = 3  
a = t1  
t2 = a  
t3 = 2  
t4 = t2 + t3  
b = t4  
t5 = a  
t6 = b  
t7 = t5 + t6  
t8 = 5  
t9 = t7 + t8  
result = t9
```